Tomorrow’s bikes

When does recent rental history help a daily forecast, and when does it fail?

Capital Bikeshare, 2011–2012
Erdős Institute Data Science Boot Camp

A forecast for the next operating day

Target: tomorrow’s total rentals across the Capital Bikeshare system

Decision context	Anticipate busy or quiet days when planning operations.
Forecast time	Just after the previous day closes, once its total is available.
Research question	How much do recent counts help, and which conditions expose their limits?

Observed rentals depend on service availability. They do not measure unmet demand.

731 daily records, with inputs kept in the past

No missing cells, repeated dates or gaps in the daily calendar

Input group	Examples	Available at forecast time?
Calendar	Weekday, holiday, season, trend	Yes
Rental history	Prior 1, 7 and 14 days; rolling averages	Yes, after daily close
Excluded	Target-day weather and rental components	No

The first 28 days supply history. 703 days remain for training and evaluation.

Learning always precedes evaluation

611 development days, followed by 92 final test days

Stage	Learn from	Evaluate on
Validation 1	Jan 2011 – Apr 3, 2012	Apr 4 – Jun 2, 2012
Validation 2	Jan 2011 – Jun 2, 2012	Jun 3 – Aug 1, 2012
Validation 3	Jan 2011 – Aug 1, 2012	Aug 2 – Sep 30, 2012
Final test	Jan 2011 – Sep 30, 2012	Oct 1 – Dec 31, 2012

Repeated daily forecasts use yesterday’s observed total. The fitted model stays fixed within each window.

A simple rule and two learned models

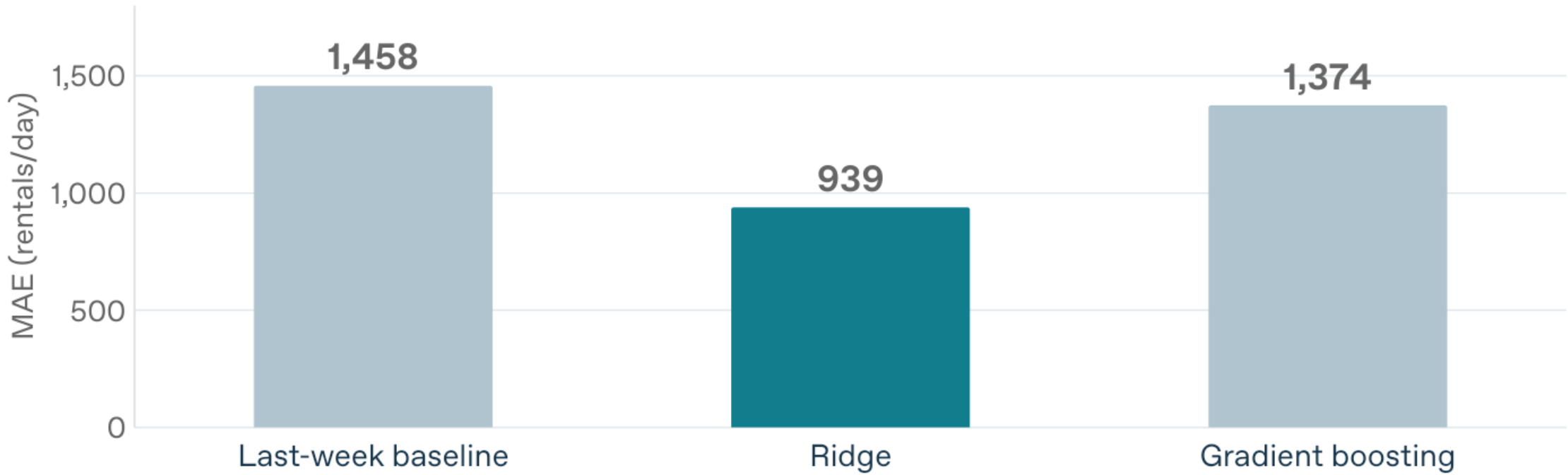
Selection metric: mean absolute error (MAE), measured in rentals per day

Last-week baseline	Predict the same count as the same weekday seven days earlier.
Ridge regression	Learn a weighted combination of calendar and recent-history clues.
Gradient boosting	Combine small decision trees to capture more flexible relationships.

Example: 6,767 predicted and 6,778 observed gives an absolute error of about 11 rentals.

Ridge reduces test MAE by 35.6%

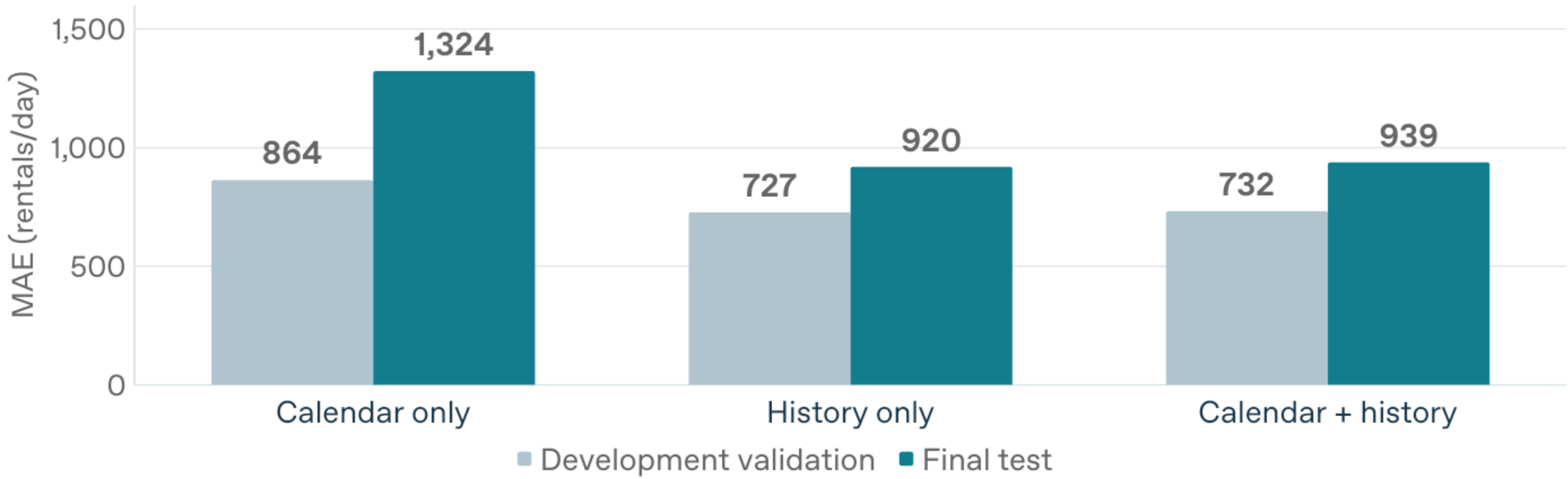
Original experiment: October–December 2012, all 92 test days retained



Lower is better. Average error remains 939 rentals/day, with some much larger misses.

Recent history supplies most of the predictive value

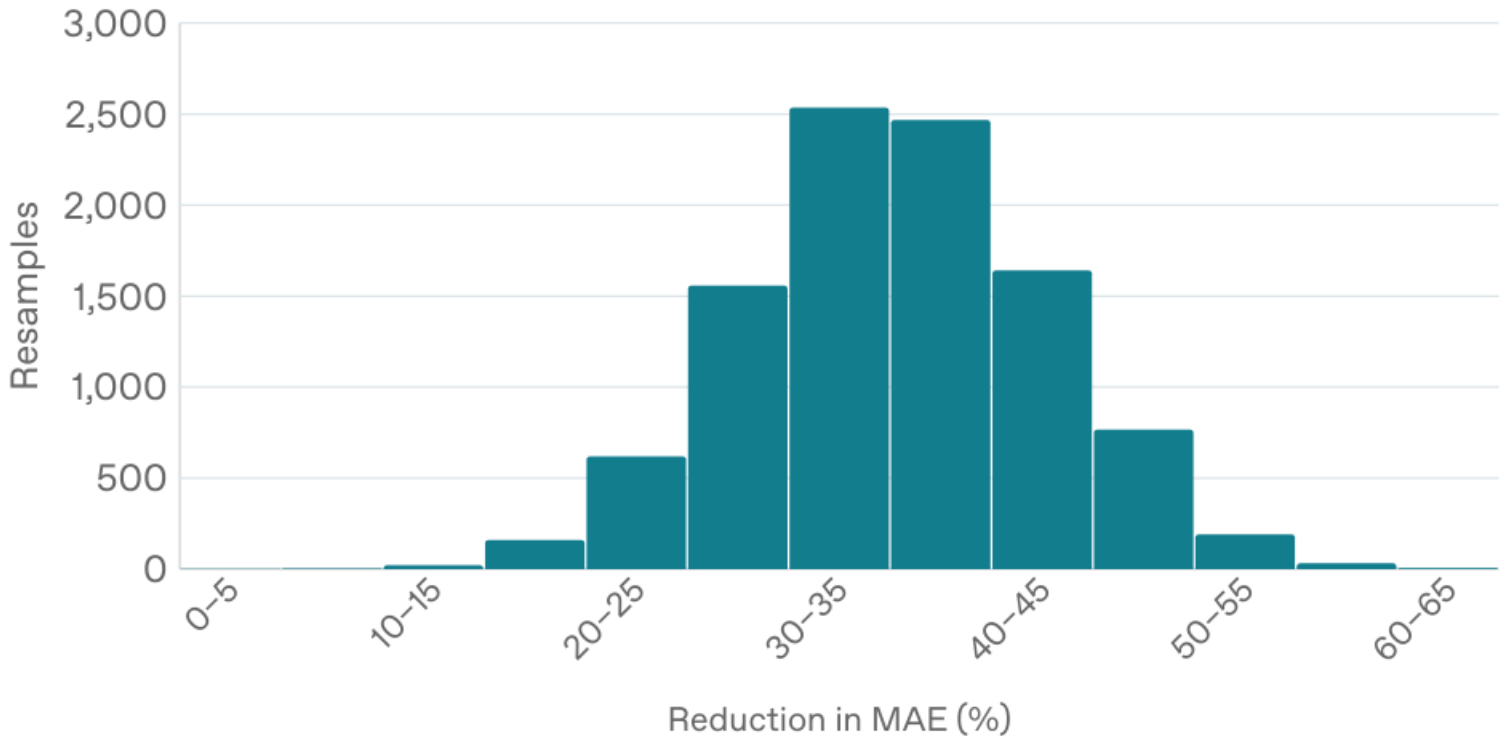
Exploratory feature comparison using the same Ridge model family



History-only and combined Ridge perform similarly. This extension does not change the original model selection.

The estimated advantage remains positive

Exploratory paired block bootstrap of the original test errors



21.1–49.7%

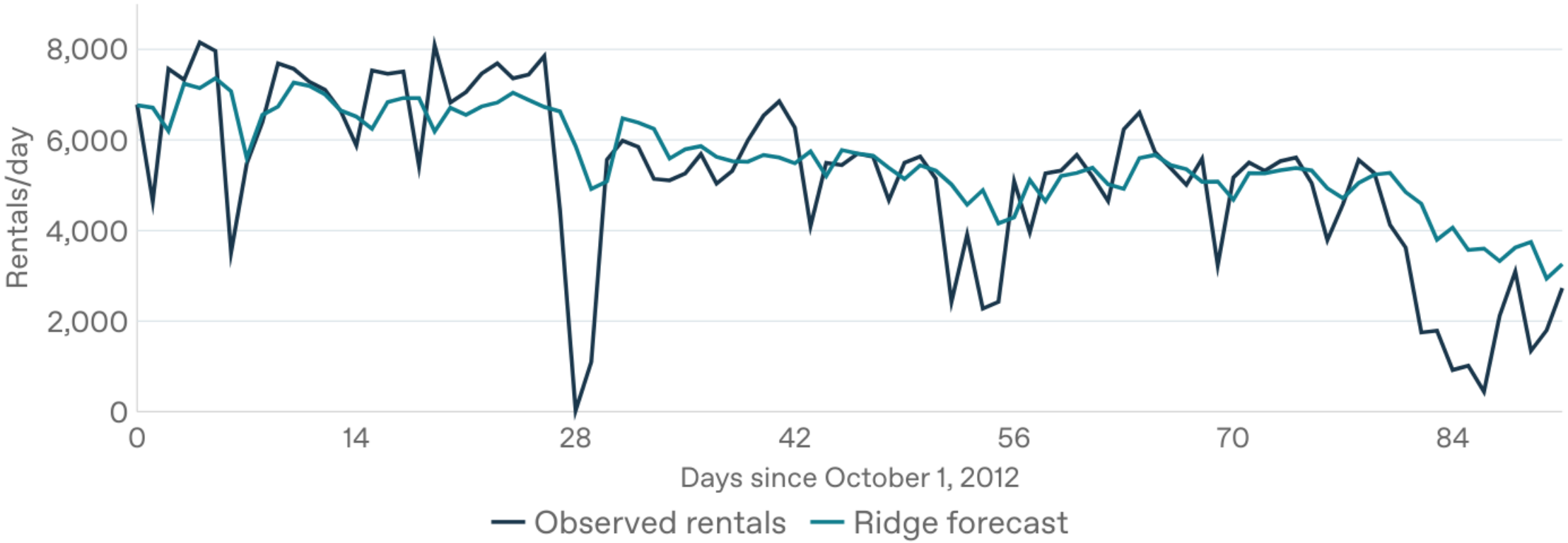
Approximate 95% interval
for reduction in MAE

7-day blocks
10,000 resamples

Conditional on this short historical period. The interval excludes uncertainty from model training and selection.

Abrupt drops expose the model’s limits

Original Ridge forecasts and observed rentals, October–December 2012



Oct 29: 5,862 predicted versus 22 recorded. Dec 26: 3,602 predicted versus 441 recorded.

Service interruptions affect recorded rentals

Post hoc case studies. All dates remain in the reported test results.

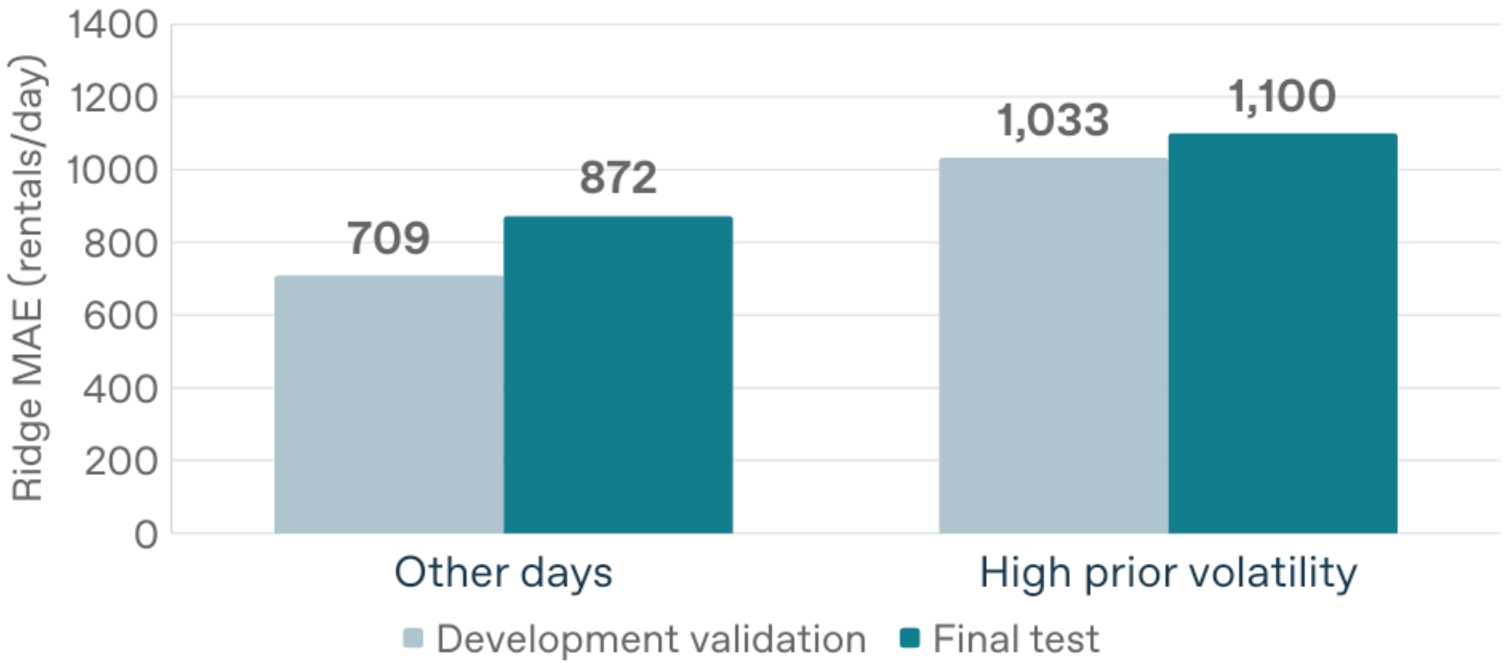
Date	Forecast / recorded	Documented context
Oct 29	5,862 / 22	Sandy shutdown, reported as about 36 hours
Dec 24	4,064 / 920	Rain in the DC metro during the holiday period
Dec 26	3,602 / 441	Winter weather in northern Virginia

Recorded rentals reflect service availability as well as customer demand.

Sources: ARLnow (Oct 31, 2012), NWS December 2012 storm report. Context does not isolate causal effects.

Recent volatility offers an imperfect warning

Exploratory flag based only on the preceding seven days



1,100 vs 872

Test MAE on flagged versus other days

27 flagged, 65 other days

The flag misses the first Sandy shutdown day. The development high-volatility group contains only 13 days.

Disruptions remain a forecasting blind spot

A simple model improves average error, while important blind spots remain

Finding	Recent rental history is useful. A simple Ridge model beats the last-week baseline.
Boundary	Abrupt closures and unusual conditions can overwhelm patterns in recent counts.
Next experiment	Add archived service notices and weather forecasts, then evaluate on new dates.

Limits: one historical system, two years, one final quarter. No claim of financial savings or station-level accuracy.

Appendix: reproducible settings and sensitivity

The notebook and project archive contain code, data and every prediction

Setting	Value
Ridge penalty candidates	0.1, 1, 10, 100 (original choice: 10)
Boosting candidates	100 or 250 trees; depth 2; learning rate 0.05
Bootstrap, 3-day blocks	95% improvement interval: 21.8%–48.5%
Bootstrap, 7-day blocks	95% improvement interval: 21.1%–49.7%
Bootstrap, 14-day blocks	95% improvement interval: 20.4%–48.7%

Exploratory extensions preserve the original predictions. Bootstrap intervals condition on the fitted models and period.

Appendix: sources and project materials

Source details also appear in the notebook and the slide notes

Data
Fanaee-T (2013), UCI Bike Sharing
doi.org/10.24432/C5W894 (CC BY 4.0)

Event context
ARLnow, October 31, 2012: Bikeshare after Sandy
National Weather Service, December 2012 storm report, pp. 2–4

Reproducible materials
Executed notebook, raw data, model code and per-day predictions
Paired block-bootstrap draws and feature-comparison results

Full source links and details of the methods appear in the slide notes.